Aqua-Loop-AI — Source Code Reference

A readable PDF reference containing the prototype's HTML, CSS, JavaScript, and README text.

Important: This PDF is for reading/reference only. To run the dashboard or submit source code to GitHub, upload the original individual files (index.html, style.css, script.js, README.md), not this PDF.

index.html

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8"><meta name="viewport" content="width=device-width,initial-scale=1">
<title>Aqua-Loop-AI | Water Intelligence Dashboard</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
  <div class="brand"><span class="drop">■</span><div><h1>Aqua-Loop-AI</h1><p>Water intelligence for data
  centres</p></div></div>
  <span class="tag">DEMO • SIMULATED DATA</span>
</header>
<main>
  <section class="intro"><div><p class="eyebrow">SUSTAINABLE DIGITAL INFRASTRUCTURE</p><h2>Make every
  drop count.</h2><p>Monitor cooling-water use, spot unusual consumption, and explore opportunities to
  reuse water.</p></div><div class="loop">■<small>MONITOR<br>→ REUSE</small></div></section>
  <section class="cards">
    <article class="metric"><p>Water used today</p><strong id="used">1,200
  <small>L</small></strong><span>Illustrative daily use</span></article>
    <article class="metric"><p>Water reused</p><strong id="reused">420
  <small>L</small></strong><span>Illustrative recovered water</span></article>
    <article class="metric"><p>Reuse share</p><strong id="share">35<small>%</small></strong><span>Reused
  ÷ water used</span></article>
  </section>
  <section class="panel">
    <div class="panel-head"><div><p class="eyebrow">WATER MONITORING</p><h3>Daily water use (sample
  data)</h3></div><button id="simulate">Simulate high-use alert</button></div>
    <div class="bars" aria-label="Illustrative daily water use chart">
      <div><span style="height:54%"></span><small>Mon</small></div><div><span
  style="height:66%"></span><small>Tue</small></div><div><span
  style="height:48%"></span><small>Wed</small></div><div><span
  style="height:72%"></span><small>Thu</small></div><div><span
  style="height:61%"></span><small>Fri</small></div><div><span
  style="height:84%"></span><small>Sat</small></div><div><span
  style="height:58%"></span><small>Sun</small></div>
    </div>
    <div id="alert" class="notice"><span>✔</span><div><strong>System status: Normal</strong><p>Sample
  readings are within the illustrative range.</p></div></div>
  </section>
  <section class="panel">
    <p class="eyebrow">PROPOSED LOOP</p><h3>How Aqua-Loop-AI could work</h3>
    <div class="steps"><div><b>1</b><strong>Monitor</strong><p>Read flow meters and tank
  levels.</p></div><div class="arrow">→</div><div><b>2</b><strong>Analyse</strong><p>Compare readings
  with normal patterns.</p></div><div class="arrow">→</div><div><b>3</b><strong>Recommend</strong><p>Flag
   unusual use and review reuse options.</p></div></div>
  </section>
  <footer>Student concept prototype • All numbers are simulated and do not represent a real data
  centre.</footer>
</main>
<script src="script.js"></script>
</body></html>
```

style.css

```
*{box-sizing:border-box}body{margin:0;background:#f1f7f7;color:#16343b;font-family:Inter,Arial,sans-
  serif}header{display:flex;justify-content:space-between;align-items:center;padding:22px
  max(5%,calc((100% - 1120px)/2));background:#fff;border-bottom:1px solid
  #dce9e8}.brand{display:flex;gap:12px;align-items:center}.drop{font-size:32px}.brand h1{font-
  size:22px;margin:0}.brand p{margin:3px 0 0;color:#698084;font-size:13px}.tag{font-size:11px;font-
  weight:800;letter-spacing:.08em;background:#e2f6ed;color:#187052;border-radius:20px;padding:9px
  12px}main{max-width:1120px;margin:28px auto;padding:0 20px}.intro{background:linear-
  gradient(120deg,#073d48,#087c78);color:white;border-radius:22px;padding:30px;display:flex;justify-
  content:space-between;align-items:center;gap:20px}.intro h2{font-size:clamp(28px,4vw,42px);margin:8px
  0}.intro p{max-width:650px;line-height:1.6;color:#e0f4f2}.eyebrow{font-size:11px;font-
  weight:800;letter-spacing:.12em;color:#2b8b7d;margin:0 0 8px}.intro
  .eyebrow{color:#a9eee0}.loop{width:110px;height:110px;border:2px solid #6de0c7;border-
  radius:50%;display:flex;flex-direction:column;align-items:center;justify-content:center;font-
  size:50px;color:#8af0d9;flex-shrink:0}.loop small{font-size:9px;line-height:1.4;text-
  align:center;letter-spacing:.08em}.cards{display:grid;grid-template-
  columns:repeat(3,1fr);gap:16px;margin:20px 0}.metric,.panel{background:white;border:1px solid
  #e0eaea;border-radius:16px;padding:22px;box-shadow:0 5px 18px #1a51500a}.metric p{margin:0 0
  14px;color:#60777b}.metric strong{font-size:32px;color:#075a60}.metric strong small{font-
  size:16px}.metric span{display:block;font-size:12px;color:#819294;margin-top:8px}.panel{margin-
  bottom:18px}.panel h3{margin:0;font-size:20px}.panel-head{display:flex;justify-content:space-
  between;align-items:center;gap:16px}button{border:0;border-
  radius:10px;background:#087c78;color:white;font-weight:700;padding:12px
  16px;cursor:pointer}button:hover{background:#075c60}.bars{height:170px;display:flex;align-
  items:end;gap:14px;padding:18px 5px 0;border-bottom:1px solid #dfe9e8;margin:10px 0 18px}.bars
  div{flex:1;height:100%;display:flex;flex-direction:column;justify-content:end;align-
  items:center;gap:8px}.bars span{display:block;width:70%;max-width:60px;background:linear-
  gradient(#47c9ad,#0a7d83);border-radius:7px 7px 0 0;min-height:10px}.bars
  small{color:#758a8c}.notice{display:flex;gap:12px;align-
  items:center;padding:13px;background:#e8f8f0;border-radius:10px;color:#176749}.notice>span{font-
  size:22px}.notice strong{font-size:14px}.notice p{margin:4px 0 0;font-
  size:12px}.notice.warning{background:#fff2dc;color:#8b4a08}.steps{display:flex;align-
  items:stretch;gap:12px;margin-top:20px}.steps>div:not(.arrow){flex:1;background:#f1f8f7;border-
  radius:12px;padding:16px}.steps b{display:flex;width:27px;height:27px;border-
  radius:50%;background:#d4f2e9;color:#087c78;align-items:center;justify-content:center;margin-
  bottom:10px}.steps strong{display:block}.steps p{font-size:13px;line-height:1.5;color:#667e80;margin-
  bottom:0}.arrow{align-self:center;color:#6ca7a1;font-size:23px}footer{text-align:center;font-
  size:12px;color:#71878a;padding:14px 0 30px}@media(max-
  width:650px){header{padding:16px;gap:8px}.tag{font-size:9px}.cards{grid-template-
  columns:1fr}.intro{padding:22px}.loop{width:76px;height:76px;font-size:35px}.steps{flex-
  direction:column}.arrow{transform:rotate(90deg)}.panel-head{align-items:flex-start;flex-
  direction:column}}
```

script.js

```
const button = document.getElementById('simulate');
const alertBox = document.getElementById('alert');
button.addEventListener('click', () => {
  alertBox.classList.add('warning');
  alertBox.innerHTML = '<span>■</span><div><strong>High water usage detected
  (simulated)</strong><p>Suggested action: inspect flow-meter readings for leaks and review whether
  suitable treated water can be reused.</p></div>';
  button.textContent = 'Alert simulated';
});
```

README.md

```
# Aqua-Loop-AI — beginner prototype

A simple browser dashboard concept for monitoring water use in data-centre cooling.

## Run it
1. Download and unzip this folder.
2. Open `index.html` in a web browser.
3. Click **Simulate high-use alert** to demonstrate the alert interaction.

## Important limitations
- This is a front-end demonstration only; it does not connect to real sensors, an AI model, or a real
  data centre.
- All figures and chart bars are illustrative simulated data.
- The alert is a scripted demonstration, not a validated leak detector.
- Real deployment would require sensor integration, engineering validation, water-quality testing,
  cybersecurity review, and expert assessment of which water streams can safely be reused.

## Files
- `index.html` — dashboard structure
- `style.css` — visual design
- `script.js` — simulated alert interaction
```